THE COMPLETE FIBRE TABLE AT $n = 41$ FOR THE TWO-TO-ONE TUNNELL MAP

AAKASH GURUNG AND KYUNGYONG LEE

The representation sets

This note accompanies the manuscript *Stable Matching and a Two-to-One Map for Tunnell's Ternary Forms*. We spell out the complete map at $n = 41$ in the original ternary coordinates. Thus

$$\mathcal{B}_{41} = \{(x, y, z) \in \mathbb{Z}^3 : 2x^2 + y^2 + 8z^2 = 41\}, \qquad \mathcal{A}_{41} = \{(u, v, w) \in \mathbb{Z}^3 : 2u^2 + v^2 + 32w^2 = 41\}.$$

There are 32 points in $\mathcal{B}_{41}$ and 16 points in $\mathcal{A}_{41}$. The tables below list every target and its two preimages, so all points of both representation sets occur.

The three rules used in the table

Even branch. For every $(u, v, w) \in \mathcal{A}_{41}$, the first preimage is

$$(u, v, 2w) \longmapsto (u, v, w).$$

This is equation (4) of the paper and is the upper bijection in the proof of Theorem 1.2.

Direct odd branch. All eight direct odd points at $n = 41$ lie in the third quarter-turn domain. Equation (10) gives

$$\Lambda_3(x, y, z) = \left(2z, y, -\frac{x}{4}\right).$$

The domain test is Proposition 3.1, and the fact that this is a bijection onto the direct target set is Theorem 3.2.

Residual odd branch. The remaining eight odd points are paired by the unique stable matching of Proposition 4.5. Equation (12) fixes the sign when the orbit matching is lifted back to individual points. Algorithm 7.1 computes the same matching, its correctness is Theorem 7.3, and the complete $n = 41$ residual run is Example 8.1. The four canonical residual pairs in original coordinates are

$$(-2, -5, -1) \longmapsto (4, 3, 0), \qquad (-2, -5, 1) \longmapsto (4, -3, 0),$$
$$(-2, 5, -1) \longmapsto (0, -3, 1), \qquad (-2, 5, 1) \longmapsto (0, -3, -1).$$

Their negatives give the other four residual pairs.

How to read a fibre

For a target $Y \in \mathcal{A}_{41}$, its *fibre* is

$$\Phi_{41}^{-1}(Y) = \{X \in \mathcal{B}_{41} : \Phi_{41}(X) = Y\}.$$

The map is designed so that this set has two elements. One has even third coordinate and comes from equation (4). The other has odd third coordinate and comes either from a quarter-turn or from the residual matching. In the two examples below, we first check that every displayed point satisfies its quadratic equation. We then determine the applicable branch and compute the image.

Date: September 2026.

Worked example 1: a direct quarter-turn fibre

Take the target

$$Y = (2, -1, 1).$$

Step 1: Check that Y is a target point. The target equation is $2u^2 + v^2 + 32w^2 = 41$. Substitution gives

$$2(2)^2 + (-1)^2 + 32(1)^2 = 8 + 1 + 32 = 41,$$

so $Y \in \mathcal{A}_{41}$.

Step 2: Find the even preimage. Equation (4) says that the even preimage of (u, v, w) is $(u, v, 2w)$. Therefore

$$X_{\text{even}} = (2, -1, 2).$$

It really is a source point because

$$2(2)^2 + (-1)^2 + 8(2)^2 = 8 + 1 + 32 = 41.$$

Its third coordinate is even, and equation (4) gives

$$\Phi_{41}(2, -1, 2) = (2, -1, 1) = Y.$$

Step 3: Check the proposed odd preimage. Consider

$$X_{\text{odd}} = (-4, -1, 1).$$

It belongs to $\mathcal{B}_{41}$, since

$$2(-4)^2 + (-1)^2 + 8(1)^2 = 32 + 1 + 8 = 41.$$

Its third coordinate is odd, so the even rule cannot apply.

Step 4: Determine the odd branch. For $X_{\text{odd}} = (x, y, z) = (-4, -1, 1)$, the first two quarter-turn tests give

$$x + 2z + y = -4 + 2 - 1 = -3, \qquad y - x + 2z = -1 + 4 + 2 = 5.$$

Neither number is divisible by 8. The third test does hold because $4 \mid x$. Proposition 3.1 shows that the three tests are mutually exclusive, so equation (10) is the applicable rule.

Step 5: Apply the quarter-turn formula. Using Theorem 3.2 and equation (10),

$$\begin{aligned}\Phi_{41}(-4, -1, 1) &= \Lambda_3(-4, -1, 1)\\ &= \left(2(1), -1, -\frac{-4}{4}\right)\\ &= (2, -1, 1) = Y.\end{aligned}$$

Step 6: Write the complete fibre. The even branch is a bijection, so Y has only one even preimage. Theorem 3.2 makes the direct odd branch a bijection onto its image, so it has only one direct odd preimage. The assembly in Theorem 1.2 shows that there are no further preimages. Hence

$$\boxed{\Phi_{41}^{-1}(2, -1, 1) = \{(2, -1, 2),\ (-4, -1, 1)\}.}$$

Worked example 2: a residual matching fibre

Take the target

$$Y = (4, -3, 0).$$

Step 1: Check that Y is a target point. We have

$$2(4)^2 + (-3)^2 + 32(0)^2 = 32 + 9 = 41,$$

so $Y \in \mathcal{A}_{41}$.

Step 2: Find the even preimage. Equation (4) gives

$$X_{\text{even}} = (4, -3, 0), \qquad \Phi_{41}(4, -3, 0) = (4, -3, 0) = Y.$$

The same calculation $2(4)^2 + (-3)^2 + 8(0)^2 = 41$ checks that this point belongs to $\mathcal{B}_{41}$.

Step 3: Check the proposed odd preimage and its branch. Consider

$$X_{\text{odd}} = (-2, -5, 1).$$

It is a source point because

$$2(-2)^2 + (-5)^2 + 8(1)^2 = 8 + 25 + 8 = 41.$$

The direct tests are

$$\begin{aligned} x + 2z + y &= -2 + 2 - 5 = -5, \\ y - x + 2z &= -5 + 2 + 2 = -1, \\ x &= -2. \end{aligned}$$

The first two values are not divisible by 8, and the last is not divisible by 4. Thus none of the three quarter-turns applies, and this point must enter the residual matching.

Step 4: Follow the residual orbit through the matching. In the lifted coordinates of Example 8.1, X_{odd} is the canonical representative

$$E = (-2, -5, 2)$$

of the source orbit s_2. Its preference order is

$$t_1,\ t_4,\ t_2,\ t_3.$$

During Algorithm 7.1, s_2 first proposes to t_1, but t_1 keeps s_1, whose edge has the smaller key. It next proposes to t_4, which keeps s_3 for the same reason. Its third proposal, to t_2, is accepted. This is the $s_2 \leftrightarrow t_2$ pair in Example 8.1. Proposition 4.5 proves that the resulting stable matching is the unique one, and Theorem 7.3 proves that the algorithm computes it.

Step 5: Recover the sign of the individual target. The canonical representative of t_2 is

$$F = (-4, 3, 0).$$

The accepted signed edge has $\eta = -1$. Since E itself was the canonical source representative, $\sigma = 1$, and equation (12) gives

$$E \longmapsto \sigma\eta F = (1)(-1)F = -F = (4, -3, 0).$$

Returning from the lifted coordinates to the original ones therefore gives

$$\Phi_{41}(-2, -5, 1) = (4, -3, 0) = Y.$$

Step 6: Write the complete fibre. Equation (4) supplies the unique even preimage. Proposition 4.5 and the sign rule (12) supply the unique residual odd preimage. Theorem 1.2 excludes any additional preimages. Therefore

$$\boxed{\Phi_{41}^{-1}(4, -3, 0) = \{(4, -3, 0),\ (-2, -5, 1)\}.}$$

A quick third example by negation

The residual construction is equivariant under negation: if $\Phi_{41}(X) = Y$, then $\Phi_{41}(-X) = -Y$. Negating the odd pair in the second example immediately gives

$$(2, 5, -1) \longmapsto (-4, 3, 0).$$

The even preimage of $(-4, 3, 0)$ is again $(-4, 3, 0)$. Hence

$$\Phi_{41}^{-1}(-4, 3, 0) = \{(-4, 3, 0),\ (2, 5, -1)\}.$$

Direct fibres

In each row, the middle point is the even preimage supplied by equation (4), and the last point is the odd preimage supplied by Theorem 3.2 and equation (10).

target $Y \in \mathcal{A}_{41}$	even preimage	odd preimage	direct computation
$(-2, -1, -1)$	$(-2, -1, -2)$	$(4, -1, -1)$	$\Lambda_3(4, -1, -1) = Y$
$(-2, -1, 1)$	$(-2, -1, 2)$	$(-4, -1, -1)$	$\Lambda_3(-4, -1, -1) = Y$
$(-2, 1, -1)$	$(-2, 1, -2)$	$(4, 1, -1)$	$\Lambda_3(4, 1, -1) = Y$
$(-2, 1, 1)$	$(-2, 1, 2)$	$(-4, 1, -1)$	$\Lambda_3(-4, 1, -1) = Y$
$(2, -1, -1)$	$(2, -1, -2)$	$(4, -1, 1)$	$\Lambda_3(4, -1, 1) = Y$
$(2, -1, 1)$	$(2, -1, 2)$	$(-4, -1, 1)$	$\Lambda_3(-4, -1, 1) = Y$
$(2, 1, -1)$	$(2, 1, -2)$	$(4, 1, 1)$	$\Lambda_3(4, 1, 1) = Y$
$(2, 1, 1)$	$(2, 1, 2)$	$(-4, 1, 1)$	$\Lambda_3(-4, 1, 1) = Y$

The sixth row is the fibre computed step by step in Worked example 1. The other seven rows use the same two formulas.

Residual fibres

The notation s_i refers to the four canonical source orbits in Example 8.1. A minus sign indicates that equation (12) applies the antipodal lift to the corresponding canonical pair.

target $Y \in \mathcal{A}_{41}$	even preimage	odd preimage	residual origin
$(-4, -3, 0)$	$(-4, -3, 0)$	$(2, 5, 1)$	negative of the s_1 pair
$(-4, 3, 0)$	$(-4, 3, 0)$	$(2, 5, -1)$	negative of the s_2 pair
$(0, -3, -1)$	$(0, -3, -2)$	$(-2, 5, 1)$	s_4 pair
$(0, -3, 1)$	$(0, -3, 2)$	$(-2, 5, -1)$	s_3 pair
$(0, 3, -1)$	$(0, 3, -2)$	$(2, -5, 1)$	negative of the s_3 pair
$(0, 3, 1)$	$(0, 3, 2)$	$(2, -5, -1)$	negative of the s_4 pair
$(4, -3, 0)$	$(4, -3, 0)$	$(-2, -5, 1)$	s_2 pair
$(4, 3, 0)$	$(4, 3, 0)$	$(-2, -5, -1)$	s_1 pair

The stable matching in Example 8.1 is obtained from the proposal sequence

$$s_1 \to t_1, \quad s_2 \to t_1 \text{ (rejected)}, \quad s_3 \to t_4,$$
$$s_4 \to t_3, \quad s_2 \to t_4 \text{ (rejected)}, \quad s_2 \to t_2.$$

It therefore matches the four source orbits as

$$s_1 \leftrightarrow t_1, \qquad s_2 \leftrightarrow t_2, \qquad s_3 \leftrightarrow t_4, \qquad s_4 \leftrightarrow t_3,$$

after which equation (12) determines the individual signs displayed in the table.

Exhaustiveness check

The direct table contains the eight targets in the image of the quarter-turn layer. The residual table contains the other eight targets. Their even preimages are the 16 points of $\mathcal{B}_{41}$ having even third coordinate; their odd preimages consist of the eight direct points and the eight residual points. Thus the tables contain 16 distinct targets and 32 distinct sources, and each target has exactly two preimages, as asserted by Theorem 1.2.

The table was also regenerated from the formalized executable map `tunnellMapTotal41`. The Lean declarations `card_BRep_41`, `card_ARep_41`, and `tunnellMapTotal41_exactly_two` verify respectively that the source and target cardinalities are 32 and 16 and that every displayed target has exactly two preimages.

Department of Mathematics, The University of Alabama, Tuscaloosa, AL 35487, USA
Email address: `agurung3@crimson.ua.edu`

Department of Mathematics, The University of Alabama, Tuscaloosa, AL 35487, USA
Email address: `klee94@ua.edu`